

# CRISTOFF STAN

## Senior Software Engineer

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## PROFESSIONAL SUMMARY

Senior Full Stack Engineer with 9+ years of experience designing and deploying interactive, cloud-native geospatial platforms using open-source technologies. Passionate about solving climate, urban, and social challenges using EO analytics, satellite imagery, and distributed mapping systems. Seeking to join a purpose-driven team building scalable, performant full-stack solutions powered by serverless architecture and modern geospatial formats.

## TECHNICAL SKILLS

- **Frontend:** JavaScript, TypeScript, React.js, MapLibre GL, DeckGL, Leaflet, Redux, HTML5, CSS3, Turf.js
- **Backend:** Node.js, Express.js, RESTful APIs, Serverless Framework, NestJS
- **Databases:** PostgreSQL + PostGIS, MySQL, MongoDB, Redis, DynamoDB
- **Geospatial:** GeoParquet, PMTiles, COG, Zarr, STAC, GDAL, QGIS, GeoJSON
- **Cloud & DevOps:** AWS (Lambda, API Gateway, S3, DynamoDB), GCP (Cloud Functions, Firestore, BigQuery), CI/CD, Docker, GitHub Actions, Terraform
- **Data & APIs:** SQL, NoSQL, OpenStreetMap, Mapbox, GraphQL, JSON, REST
- **Others:** Agile/Scrum, Git, Jira, VSCode, Jest, Cypress, Playwright

## WORK EXPERIENCE

### Senior Software Engineer

#### Sensidev

September 2021 – December 2024

Timișoara, Romania (Remote)

- Designed and deployed highly interactive geospatial frontends using React, DeckGL, and MapLibre GL to visualize socio-economic and EO datasets in real-time.
- Architected cloud-native APIs using Node.js and Serverless Framework hosted on AWS Lambda and GCP Cloud Functions to serve scalable data-driven applications.
- Built client-facing tools enabling satellite data interaction, tile streaming (PMTiles/COG), and multi-resolution raster handling using modern open formats.
- Developed backend services supporting STAC-compliant catalog integration for satellite imagery search and distribution workflows.
- Engineered complex spatial queries in PostgreSQL + PostGIS to analyze urban density, mobility patterns, and climate impacts using mobile + EO data fusion.

- Integrated map tile services and vector overlays from OSM, Mapbox, and custom layers for dynamic rendering and analytics interpretation.
- Collaborated across distributed teams in Europe to build and deploy sustainable development monitoring dashboards with automated CI/CD pipelines.
- Optimized front-end rendering using React virtual DOM techniques for massive raster/vector datasets with DeckGL/MapLibre hybrid visualizations.
- Led unit, integration, and e2e testing efforts (Jest, Cypress) to ensure high test coverage for mission-critical spatial intelligence tools.

## **Software Engineer**

### **Syscom Digital**

**October 2019 – August 2021**

Bucharest, Romania (Remote)

- Built REST APIs and full-stack dashboards to visualize real-time event geolocation data using Leaflet and PostgreSQL/PostGIS.
- Designed microservices using Node.js and deployed containerized services on AWS with Docker and GitHub Actions for automated build and release.
- Developed user-facing UIs for public service data interaction, integrating OGC/WMS/WFS protocols for dynamic map layers and overlays.
- Integrated geocoding and reverse-geocoding workflows with external APIs to enrich user-input spatial data points.
- Applied spatial joins and raster overlays to generate mobility heatmaps for advertising impact analytics.
- Implemented indexed search for large vector datasets using GeoJSON and spatial trees for performance.
- Ensured mobile-first responsive designs using custom CSS and performance optimization techniques.
- Collaborated with marketing and analytics teams to build tools for real-time geospatial campaign visualization and ROI prediction.

## **Software Developer**

### **Soft Build**

**June 2016 – July 2019**

Timișoara, Romania (Remote)

- Designed backend APIs for geospatial data ingestion pipelines using Node.js and MongoDB for NoSQL spatial document storage.
- Built custom raster tile generators using GDAL and Python for rendering EO datasets as slippy maps.
- Integrated client web apps with OGC-compliant WMS/WFS geoservers to deliver dynamic satellite imagery overlays.
- Managed schema migrations and relational data operations in PostgreSQL with a focus on spatial indexing.
- Developed internal map tools allowing annotations, polygon drawing, and layer toggles for internal GIS teams.
- Maintained reusable React components to support thematic map styles, layer legends, and analytics panels.
- Leveraged AWS services (S3, Lambda, API Gateway) for high-availability public data API endpoints.
- Designed and executed tests for all geospatial rendering services to ensure cross-browser compatibility.

- Participated in system design and architecture discussions for building modular geospatial web platforms.
- Delivered documentation and usage guides for internal and client-side consumption of mapping APIs.
- Implemented data versioning and ETL workflows to support incremental updates from external EO data providers.
- Managed error handling, logging, and observability using centralized logging and distributed tracing.

## **Software Developer**

### **Decima Digital**

**November 2015 – April 2016**

Tallinn, Estonia

- Built modular React.js components to visualize real estate and property-based geospatial data.
- Developed and deployed microservices for spatial queries and reverse lookups using MongoDB and custom grid-based logic.
- Maintained responsive map applications using Leaflet and custom raster tile sets from internal servers.
- Refactored front-end visualization code to improve load times for large shapefiles and raster tiles.
- Supported the migration from legacy mapping frameworks to modern JavaScript libraries.
- Automated unit tests using Mocha and integrated into CI pipeline for regression testing.
- Collaborated with frontend designers to translate UI wireframes into interactive map-based dashboards.
- Conducted performance audits and implemented best practices for front-end map rendering efficiency.

## **Software Developer Intern**

### **Borna Technology**

**September 2014 – August 2015**

Tallinn, Estonia (On-Site)

- Created simple geospatial dashboards using JavaScript and HTML5 Canvas for real-time sensor feeds.
- Supported senior engineers with development of PostgreSQL queries for early urban mapping experiments.
- Maintained code quality and participated in code reviews to understand enterprise-grade geospatial solutions.
- Learned integration techniques for spatial APIs and web-based data visualizations.
- Assisted in testing mobile UI versions of map products across various devices.
- Contributed to basic API documentation for internal location-based data services.
- Participated in cross-functional team meetings and user testing sessions for map tool feedback.
- Improved internal tool performance for querying address-based datasets using bounding box optimizations.

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## **EDUCATION**

### **Tallinn University**

Bachelor's Degree in Computer Science,  
**2010 – 2014**

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## CERTIFICATIONS

- ✓ AWS Certified Solutions Architect – Associate
  - ✓ Google Cloud Associate Cloud Engineer
  - ✓ PostGIS Spatial Data Certification – Udemy
  - ✓ Full-Stack Web Development Specialization – Coursera
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## PROJECTS

### Urban Climate Insight Tool

- Built a web application for dynamic climate risk visualization using DeckGL, MapLibre GL, and PMTiles from EO sources.
- Leveraged AWS Lambda and PostgreSQL/PostGIS to process and serve city-scale analytics through interactive maps.

### Satellite Data STAC Catalog Browser

- Created a frontend tool to query, filter, and display STAC-compliant EO metadata and assets.
- Developed Node.js backend to serve JSON metadata and COG links using a fast, cache-first architecture.

### Distributed Disaster Monitoring Dashboard

- Developed serverless API + UI to aggregate mobile and satellite feeds for natural disaster tracking in real-time.
- Designed advanced map interactions using geospatial overlays and time-based data layers.